WORKED SOLUTION

STEP III 2014, Question 4

Pure • Integral Inequalities

STEP III 2014, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q4

This question has a full guided treatment in *Beyond STEP*, Section 4.7, *The Calculus of Variations*: https://beyondstep.keithmcnulty.org/ch04/variations.html

The key idea

Expand the square in I_1 and compare it with I: the difference is the integral of an exact derivative, $\frac{\mathrm{d}}{\mathrm{d}x}\left(y^2 \tan x\right)$, which vanishes because $y(1) = 0$ and $\tan 0 = 0$. Once I is written as the integral of a square it is obviously non-negative. In part (ii) the same trick works provided we choose b so that the $\sec^2$ term produces exactly a^2y^2, and the only thing that can go wrong is $\tan bx$ blowing up inside $[0,1]$.

Solution

Throughout we assume, as the question intends, that y has a continuous derivative on $[0,1]$.

Part (i)

Since $0 \leq x \leq 1 < \frac{\pi}{2}$, the function $\tan x$ is continuous on $[0,1]$, so I_1 makes sense. Expanding the square,

$$I_1 = \int_0^1 \left((y')^2 + 2yy' \tan x + y^2 \tan^2 x\right) \mathrm{d}x,$$

so

$$\begin{aligned} I - I_1 &= -\int_0^1 \left(y^2 + 2yy' \tan x + y^2 \tan^2 x\right) \mathrm{d}x \\ &= -\int_0^1 \left(2yy' \tan x + y^2 \sec^2 x\right) \mathrm{d}x, \end{aligned}$$

using $1+\tan^2 x = \sec^2 x$. By the product rule, $\frac{\mathrm{d}}{\mathrm{d}x}(y^2\tan x) = 2yy'\tan x + y^2\sec^2 x$, which is exactly the integrand. Hence

$$I - I_1 = -\Big[y^2\tan x\Big]_0^1 = -\big(y(1)^2\tan 1 - y(0)^2\tan 0\big) = 0,$$

because $y(1)=0$ and $\tan 0 = 0$. So $I = I_1$, as required.

Since I_1 is the integral of a square over an interval of positive length, $I_1 \geq 0$, and therefore $I \geq 0$.

Now suppose $I = 0$. Then $I_1 = 0$, and the integrand $(y' + y\tan x)^2$ is continuous and non-negative, so it must vanish everywhere on $[0,1]$ (if it were positive at one point it would be positive on a small interval, making the integral positive). Thus $y' + y\tan x = 0$ for $0 \leq x \leq 1$.

Ask yourself How do we solve $y' + y\tan x = 0$ without dividing by y, which might be zero somewhere?

Use an integrating factor. Since $\cos x > 0$ on $[0,1]$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\cos x}\right) = \frac{y'\cos x + y\sin x}{\cos^2 x} = \frac{y' + y\tan x}{\cos x} = 0,$$

so $y = C\cos x$ for some constant C. The condition $y(1) = 0$ gives $C\cos 1 = 0$, and $\cos 1 \neq 0$, so $C = 0$. Hence $I = 0$ only if $y = 0$ for all x in $[0,1]$.

Part (ii)

Let $K = \displaystyle\int_0^1 (y' + ay\tan bx)^2\,\mathrm{d}x$, with $b > 0$ still to be chosen. Expanding,

$$K = \int_0^1 \left((y')^2 + 2ayy'\tan bx + a^2y^2\tan^2 bx\right)\mathrm{d}x.$$

The derivative we hope to see is

$$\frac{\mathrm{d}}{\mathrm{d}x}(ay^2\tan bx) = 2ayy'\tan bx + aby^2\sec^2 bx = 2ayy'\tan bx + aby^2 + aby^2\tan^2 bx.$$

Comparing this with the integrand of K, the $\tan^2$ terms match exactly when $ab = a^2$, that is, $b = a$. With this choice,

$$(y' + ay\tan ax)^2 = (y')^2 - a^2y^2 + \frac{\mathrm{d}}{\mathrm{d}x}(ay^2\tan ax),$$

and so

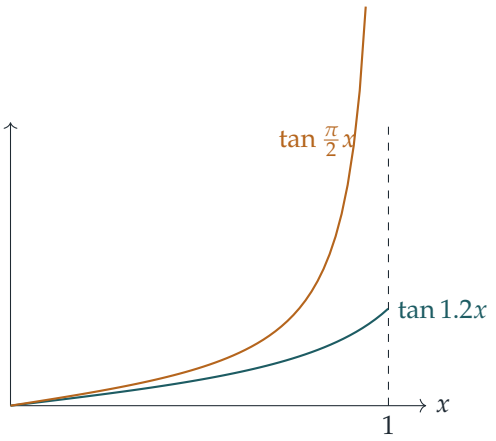
$$K = J + \Big[ay^2\tan ax\Big]_0^1 = J + a\,y(1)^2\tan a = J,$$

since $y(1) = 0$ and $\tan 0 = 0$.

This argument needs $\tan ax$ to be defined and continuous on the whole of $[0,1]$, so that K exists and the fundamental theorem of calculus applies. As $a>0$, ax runs over $[0,a]$, and tan is continuous there exactly when $a<\frac{\pi}{2}$. So the proof is valid for

$$a<k=\frac{\pi}{2}.$$

For such a we have $J=K\geq 0$, since K is the integral of a square. (In fact, the argument of part (i) with $\cos ax$ in place of $\cos x$ shows that $J=0$ would force $y=C\cos ax$ with $C\cos a=0$, so $y\equiv 0$; for y not identically zero we even get $J>0$.)



The picture shows why $k=\frac{\pi}{2}$: for $a<\frac{\pi}{2}$ the weight $\tan ax$ stays bounded on $[0,1]$, but at $a=\frac{\pi}{2}$ it has an asymptote at the end point $x=1$.

The case $a=\frac{\pi}{2}$

The proof suggests where to look: $K=0$ needs $y'+ay\tan ax=0$, whose solutions are $y=C\cos ax$. For $a<\frac{\pi}{2}$ these fail the condition $y(1)=0$, but for $a=\frac{\pi}{2}$ they satisfy it. So try

$$y=\cos\tfrac{\pi x}{2},\qquad y'=-\tfrac{\pi}{2}\sin\tfrac{\pi x}{2},$$

which is not everywhere zero and has $y(1)=\cos\frac{\pi}{2}=0$. Then

$$\begin{aligned}J&=\frac{\pi^2}{4}\int_0^1\left(\sin^2\tfrac{\pi x}{2}-\cos^2\tfrac{\pi x}{2}\right)\mathrm{d}x=-\frac{\pi^2}{4}\int_0^1\cos\pi x\,\mathrm{d}x\\&=-\frac{\pi^2}{4}\left[\frac{\sin\pi x}{\pi}\right]_0^1=0.\end{aligned}$$

So $y=\cos\frac{\pi x}{2}$ (or any non-zero multiple of it) gives $J=0$ when $a=k$.

Checks

- Take $y=1-x$ in part (i): $I=\int_0^1\left(1-(1-x)^2\right)\mathrm{d}x=1-\frac{1}{3}=\frac{2}{3}>0$, and evaluating I_1 numerically also gives 0.6667.

- For many random cubic multiples of $(1-x)$ and random a in $(0, \frac{\pi}{2})$, a computer check confirmed $J = K > 0$ to high accuracy.
- The bound $k = \frac{\pi}{2}$ is sharp: with $y = \cos\frac{\pi x}{2}$ one finds $J = \frac{1}{2}\left(\frac{\pi^2}{4} - a^2\right)$, which is negative for every $a > \frac{\pi}{2}$ (for example $J \approx -0.766$ when $a = 2$).

Common slips

- Forgetting that the boundary term at $x = 0$ vanishes because $\tan 0 = 0$, not because of any condition on $y(0)$; nothing is assumed about $y(0)$.
- Concluding $y' + y\tan x = 0$ from $I_1 = 0$ without saying why: it is the continuity of the integrand that matters.
- Choosing b without checking that $\tan bx$ stays finite on $[0,1]$; that is exactly where the condition $a < \frac{\pi}{2}$ comes from.

Beyond STEP

The inequality $\int_0^1 (y')^2\,\mathrm{d}x \geq \frac{\pi^2}{4}\int_0^1 y^2\,\mathrm{d}x$ for $y(1) = 0$ is a form of Wirtinger's (or Poincaré's) inequality, and the constant $\frac{\pi^2}{4}$ is the smallest eigenvalue of $-y'' = \lambda y$ with $y'(0) = 0$, $y(1) = 0$; the extremal function $\cos\frac{\pi x}{2}$ is its eigenfunction. In the calculus of variations, the trick of completing the square with a well-chosen derivative is the idea behind Legendre's and Jacobi's tests for whether a stationary path really is a minimum. For more questions in this spirit, see Siklos, *Advanced Problems in Mathematics*; for the underlying eigenvalue problems, Boyce and DiPrima, *Elementary Differential Equations*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.